

EXP NO. 9**Develop an Android Calculator Application using
Android Studio****AIM:**

To develop an Android application that implements a basic calculator using Android controls such as Button, TextView, and EditText. The calculator will support basic mathematical operations like Addition, Subtraction, Multiplication, and Division.

ALGORITHM:**1. Initialize the Android Application:**

- o Set up a new Android project in Android Studio.
- o Create a MainActivity Java or Kotlin file for the main screen of the application.

2. Design the User Interface (UI):

- o Create the following UI components in the XML layout file:
 - ☐ EditText for input fields where the user will enter numbers (two EditTexts for number1 and number2).
 - ☐ Buttons for each arithmetic operation (Addition, Subtraction, Multiplication, Division).
 - ☐ TextView to display the result of the operation.

3. Declare Variables:

- o Create variables for the EditText fields, Buttons, and TextView.
- o Define variables to hold the input numbers and the result (e.g., num1, num2, result).

4. Get Input from the User:

- o Get the values entered by the user in EditText fields using `getText().toString()` and convert them to numerical values (like `Integer.parseInt()` or `Double.parseDouble()`).

5. Set Event Listeners for Buttons:

- o For each button (Addition, Subtraction, Multiplication, and Division), set an `OnClickListener`.
- o Inside the `onClick()` method for each button, perform the respective arithmetic operation:
 - ☐ Addition: $\text{result} = \text{num1} + \text{num2}$

- ❑ Subtraction: $\text{result} = \text{num1} - \text{num2}$
- ❑ Multiplication: $\text{result} = \text{num1} * \text{num2}$
- ❑ Division: $\text{result} = \text{num1} / \text{num2}$ (add validation to prevent division by zero).

6. Perform Arithmetic Operation:

- o Check which button was clicked (using `view.getId()`), and based on the button, perform the respective arithmetic operation.
- o Handle Division by Zero: Check if `num2` is 0 before performing division and show an appropriate error message if true.

7. Display the Result:

- o Use `TextView` to display the result of the operation.
- o Convert the result to a string and set it using `setText()` method on the `TextView`.

8. Validate User Inputs:

- o Before performing any operation, validate the input to ensure the user has entered valid numbers (not empty or non-numeric characters).

9. Handle Edge Cases:

- o Ensure the app handles cases where users input invalid values or attempt division by zero. Display appropriate messages for errors.

10. Finish the Operation:

- o Once the result is displayed, the user can either perform another calculation or exit the app.

CODE:

ACTIVITY.XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

<!-- Input 1 -->

<EditText

android:id="@+id/num1"
android:hint="Enter first number"
android:inputType="numberDecimal"
android:layout_width="match_parent"
android:layout_height="wrap_content"/>

<!-- Input 2 -->

<EditText

android:id="@+id/num2"
android:hint="Enter second number"
android:inputType="numberDecimal"
android:layout_width="match_parent"
android:layout_height="wrap_content"/>

<!-- Buttons -->

<LinearLayout

android:orientation="horizontal"
android:layout_width="match_parent"
android:layout_height="wrap_content">

<Button android:id="@+id/add" android:text="+" android:layout_weight="1"
android:layout_width="0dp" android:layout_height="wrap_content"/>

<Button android:id="@+id/sub" android:text="-" android:layout_weight="1"
android:layout_width="0dp" android:layout_height="wrap_content"/>

<Button android:id="@+id/mul" android:text="*" android:layout_weight="1"
android:layout_width="0dp" android:layout_height="wrap_content"/>

<Button android:id="@+id/div" android:text="/" android:layout_weight="1"
android:layout_width="0dp" android:layout_height="wrap_content"/>

</LinearLayout>

<!-- Result -->

```
<TextView
    android:id="@+id/result"
    android:text="Result"
    android:textSize="24sp"
    android:layout_marginTop="20dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```

MAIN_ACTIVITY.JAVA

```
package com.example.calculatorapp;

import android.os.Bundle;
import android.view.View;
import android.widget.*;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText num1, num2;
    Button add, sub, mul, div;
    TextView result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        num1 = findViewById(R.id.num1);
        num2 = findViewById(R.id.num2);
        add = findViewById(R.id.add);
```

```

sub = findViewById(R.id.sub);
mul = findViewById(R.id.mul);
div = findViewById(R.id.div);
result = findViewById(R.id.result);

add.setOnClickListener(v -> calculate('+'));
sub.setOnClickListener(v -> calculate('-'));
mul.setOnClickListener(v -> calculate('*'));
div.setOnClickListener(v -> calculate('/'));
}

private void calculate(char operator) {
    String s1 = num1.getText().toString();
    String s2 = num2.getText().toString();

    if (s1.isEmpty() || s2.isEmpty()) {
        result.setText("Enter both numbers");
        return;
    }

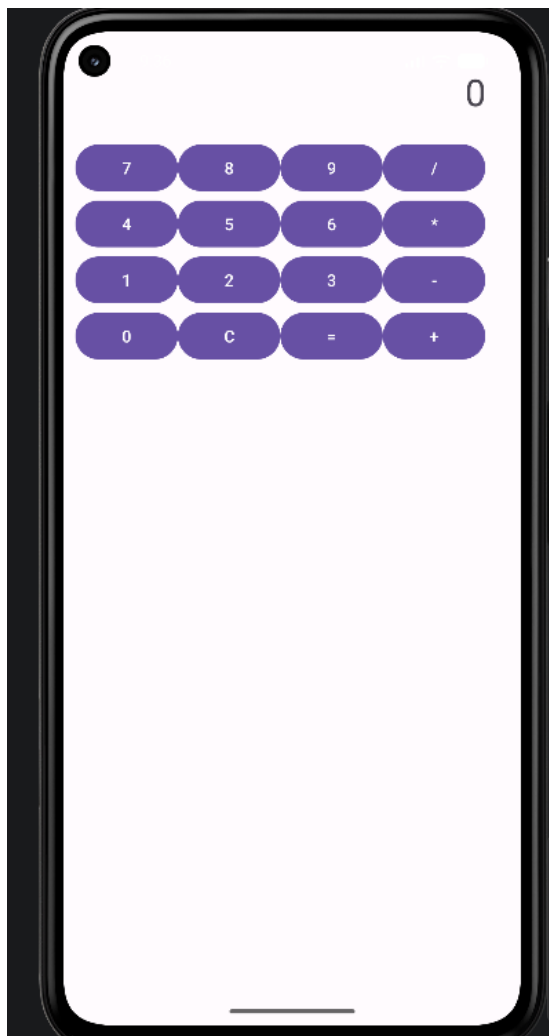
    double n1 = Double.parseDouble(s1);
    double n2 = Double.parseDouble(s2);
    double res = 0;

    switch (operator) {
        case '+': res = n1 + n2; break;
        case '-': res = n1 - n2; break;
        case '*': res = n1 * n2; break;
        case '/':
            if (n2 == 0) {
                result.setText("Cannot divide by zero");
                return;
            }

```

```
    }  
    res = n1 / n2;  
    break;  
}  
  
    result.setText("Result: " + res);  
}  
}
```

OUTPUT:



RESULT:

A calculator is designed using Android studio which uses controls like Button, TextView, EditText having basic functionality like Addition, Subtraction, multiplication, and Division.